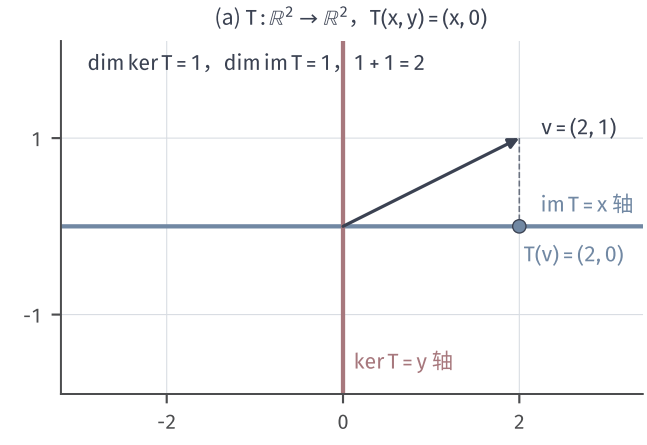
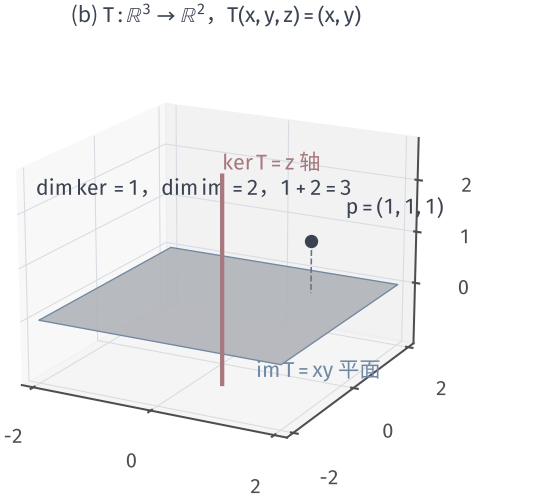

(a) T : ℝ² → ℝ²， T(x, y) = (x, 0)
dim ker T = 1， dim im T = 1， 1 + 1 = 2
v = (2, 1)
im T = x 轴
T(v) = (2, 0)
ker T = y 轴
1
-1
-2
0
2


(b) T : ℝ³ → ℝ²， T(x, y, z) = (x, y)
ker T = z 轴
dim ker = 1， dim im = 2， 1 + 2 = 3
p = (1, 1, 1)
im T = xy 平面
2
1
0
-2
0
2
2
0
-2